Evaluating Practical Enumeration and Blocking Attacks on the Snowflake Circumvention System

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A Artifact Appendix

A.1 Abstract

This artifact contains source code, experiment dockers, compressed logs, result analysis notebooks, and paper-ready figures for studying enumeration and blocking of Tor Snowflake proxies. Docker Compose builds two step-loop simulators and an opt-in real-world prober. The simulators produce per-scenario logs and attacker observations; the notebooks turn bundled logs and measurement data into PDF figures.

A.2 Description & Requirements

The root `README.md` is the operational guide. `compose.yaml` and `Dockerfile` define all containers. `general-simulation/` contains the enumeration/blocking simulator, scenario scripts, compressed results, `Selected_Simulation_Figures.ipynb`, and its plotting helper. `malicious-proxy-simulation/` contains the malicious-proxy variant and bundled logs. `real-world/` contains the modified Snowflake client, analysis code, `Real_World_Selected_Figures.ipynb`, ASN metadata, and checked-in figures. `prefix-data/` contains IPv4/IPv6 prefix-to-ASN data used by the live prober.

A.2.1 Security, privacy, and ethical concerns. Firstly, real-world IP addresses are not shared in the artifact. Readers are expected to run `real-world` by themselves but needs to obey the rate limit set by us and avoid causing pressure to live Snowflake system. Observed IP addresses are mapped to ASNs and anonymized with a per-run keyed HMAC-SHA256 value. Raw IPs and hashes from the study are withheld to protect volunteer operators, and hashes from separate runs cannot be correlated. For simulation, the simulations are local and use synthetic entities. All the users and proxies are simulated and don't leak real user or proxy information.

A.2.2 How to access. The archived source is at https://zenodo.org/records/23776345. The aggregate archive expected by the real-world notebook, `real-world/data/real-results.zip`, is not included in this checkout for security concerns. The github repo is also maintained at https://github.com/r-andlab/snowflake-enumeration

A.2.3 Hardware dependencies. No special CPU, GPU, FPGA, or hardware counter is required. A five-simulated-day run is a 12-core server with 70 GB RAM, taking approximately one compute-hour; smaller systems can reduce `SIM_DAYS`, `EPISODES`, or scenario scope.

A.2.4 Software dependencies. Recommended: Docker Engine with Compose v2, and Internet access for the initial image build. The image builds the Go modules with Go 1.23. Plotting requires Python 3, Jupyter, pandas, NumPy, and Matplotlib. Dependency versions for the Python notebooks are not pinned. Local, non-container builds require Go 1.23 or newer.

A.2.5 Benchmarks. Simulation inputs include hourly proxy-population traces embedded in the Go source. Bundled `sim*.log.gz` files cover baseline, attacker-fraction, churn, connection-duration, defense, standalone-churn, and malicious-proxy settings. The live prober uses the RouteViews prefix-to-ASN files in `prefix-data/`. Real-world plotting additionally requires the absent `real-results.zip` plus the included `asn_info.zip`; raw identifying measurements are intentionally not distributed.

A.3 Set Up

A.3.1 Installation. Download the artifact archive from the Zenodo record, verify its checksum against the value displayed by Zenodo, extract it, enter the extracted repository root, and build the image:

```
# Download the artifact file from:
# https://zenodo.org/records/23776345
unzip <downloaded-artifact>.zip
cd <extracted-artifact-directory>
docker compose build
```

For plotting, create an environment and install `jupyter`, `pandas`, `numpy`, and `matplotlib`. Do not enable the `live` profile for simulation-only evaluation.

A.3.2 Basic test. First validate Compose, then run a short default enumeration episode:

```
docker compose config --quiet
docker compose run --rm --no-deps \
  -e SIM_DAYS=0.01 -e EPISODES=1 \
  -e PROBER_START_HOURS=0 \
  -e ONLY_REGEX='^default$' \
  general-simulation \
  /bin/bash -c './run_enumeration.sh'
```

Success is an exit status of zero and messages naming `/output/default/enumeration/episode-1/sim.log` and stating that all requested scenarios completed. Inspect it with the tail command in

the root README. The short run checks functionality only and is not scientifically representative.

A.4 Evaluation Workflow

A.4.1 Major claims.

(C1): *Takeaway 1 (§5.1.1):* Proxy churn prevents complete, sustained IP enumeration, but modest probing quickly reveals stable, high-capacity proxies that are disproportionately important for blocking.

(C2): *Takeaway 2 (§5.1.1):* The June 2025 Iran event shifted the attacker's view toward a smaller, more stable, infrastructure-heavy subset with large capacity.

(C3): *Takeaway 3 (§5.1.1):* A small number of countries, ASes, and AS types account for a large fraction of enumerated proxies.

(C4): *Takeaway 4 (§5.1.2):* Churn makes IP-level blocking disruptive but manageable, whereas AS-level blocking can drive client retry burdens into the thousands.

(C5): *Takeaway 5 (§5.1.2):* More than 50% of observed proxies can be blocked with negligible Tranco Top-100 collateral because they are in ISP networks, but collateral rises sharply when major cloud ASes are added.

(C6): *Takeaway 6 (§5.2.1):* Enumeration success is driven primarily by attacker scale; many aggressive pollers reach high coverage even under high proxy churn.

(C7): *Takeaway 7 (§5.2.2):* More attackers, lower churn, and shorter sessions make blocking extremely costly, while higher churn reduces harm and remains Snowflake's primary defense against IP blocking.

(C8): *Takeaway 8 (§5.2.3):* Slower standalone-proxy churn reduces attackers' enumeration coverage but increases client retries under blocking.

(C9): *Takeaway 9 (§6):* One restricted and one unrestricted malicious proxy polling aggressively received an average of 19.8% of all client assignments over 15 simulated days.

The claim-to-experiment mapping is: C1–C3 ← E1; C4–C5 ← E2; C6–C8 ← E3; and C9 ← E4.

A.4.2 Experiments.

(E1): *Real-world enumeration, blocking, and collateral [proves C1–C5].* **Preparation:** This experiment contacts the public Snowflake broker; obtain authorization and apply the paper's bounded-measurement safeguards. The two services advertise different NAT labels and store results in separate volumes. From the artifact root, start and monitor them as in the README:

```
docker compose --profile live up \
 --build -d real-world \
 real-world-restricted
docker compose --profile live ps --all \
 real-world real-world-restricted
docker compose --profile live logs \
 --follow --tail=50 real-world \
 real-world-restricted
```

Let the bounded experiment run for at least five days, as recommended by the README; exact reproduction of the paper's counts uses its authorized 48-day measurement archive. Stop the services with

```
docker compose --profile
  live stop real-world real-world-restricted
```

, then copy each CSV before removing its container:

```
docker compose --profile live cp \
 real-world:/output/proxy_ASNs.csv \
 ./unrestricted.csv
docker compose --profile live cp \
 real-world-restricted:/output/\
proxy_ASNs.csv ./restricted.csv
```

As directed by the README, use `real-world/asn_info.zip` for ASN/domain metadata and run `Real_World_Selected_Figures.ipynb`. For the exact paper figures, place the authorized `real-results.zip` in `real-world/data/`; the current notebook reads both ZIP archives directly. Execute it entirely in Docker:

```
docker run --rm -e HOME=/tmp \
 --user "$(id -u):$(id -g)" \
 -v "$PWD:/artifact" \
 -w /artifact/real-world \
 python:3.12-bookworm sh -lc '
 python -m venv /tmp/ae &&
 /tmp/ae/bin/pip install -q \
  jupyter pandas numpy matplotlib &&
 /tmp/ae/bin/jupyter nbconvert \
  --to notebook --execute --inplace \
  Real_World_Selected_Figures.ipynb'
```

Results: The notebook emits five PDFs. The enumeration/retention figures establish C1–C3. The three blocking figures show that ASN blocking drives retries into the thousands, the top 1% of observed ASes covers over 30% of proxies with 0% Top-100 collateral, and collateral rises at major cloud ASes, establishing C4–C5.

(E2): *General enumeration and blocking simulation [proves C6–C8].* **Preparation and execution:** From the artifact root, use Compose for the paper's 30-day, three-episode campaign, then copy its volume before removing the container:

```
SIM_DAYS=30 EPISODES=3 \
 docker compose up --build \
 general-simulation
mkdir -p "general-simulation/data/\
Simulation Results"
docker compose cp \
 general-simulation:/output/. \
 "general-simulation/data/\
Simulation Results/"
```

Execute `Selected_Simulation_Figures.ipynb` with the same `dockerrun` recipe as E1, changing the working directory to `/artifact/general-simulation` and the notebook filename accordingly.

Results: The inventory must select the new logs and export 12 PDFs. Attacker scale dominates enumeration; more attackers, lower churn, and shorter sessions worsen blocking; higher churn reduces harm; and bundled standalone-churn

runs show the enumeration/retry tradeoff. Together these results prove C6–C8.

(E3): *Malicious-proxy simulation [proves C9].* **Execution:** Run the 15-day case study entirely through Compose:

```
SIM_DAYS=15 EPISODES=1 \
 docker compose up --build \
 malicious-proxy-simulation
```

Inspect the output volume with the Docker tail command in the README. For every `minute-malicious-proxy` record, add the two `pct_match_events_vs_new_connections` values and average across all 21,600 minutes; this calculation can be performed by mounting the copied log into the Python container used in E1.

Results: The result of bundled runs should be consistent with the paper's 19.8% mean and proving C9.

A.5 Notes on Reusability

Simulation experiments can be scaled with `SIM_DAYS`, `EPISODES`, `ONLY_REGEX`, and the documented `SNOWFLAKE_SIM_*` variables. The scripts expose attacker population, client population, churn, NAT mix, polling intervals, connection duration, and broker timeout. New logs following the documented scenario/mode/episode layout are discovered automatically by the plotting helper. The standalone half-churn Compose service illustrates adding a dedicated variant without altering the baseline datasets.

A.6 Version

Based on the LaTeX template for Artifact Evaluation V20220926.